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Research question. Can an educator, controlling only the assignment text itself, seed a watermark that makes a verbatim LLM copy-paste expose itself?

■ Contributions

1. **Input-side watermarking, formalized** for academic integrity, with a threat model whose adversary is a student pasting the brief verbatim into an LLM.
2. **SteganoPrompt** — a single-file, zero-dependency web tool that hides any ASCII payload in the invisible Unicode Tags block `U+E0000–U+E007F`.
3. Evaluation across **eight LLM families** and the content channels of real classroom workflows (Word, Google Docs, PDF, e-mail, Slack, LMS editors).
4. **Disclosure built in:** every encoded brief carries an always-on integrity reminder, plus limitations, defences and concrete guidance for ethical classroom use.

■ Evaluation Protocol

Compliance bins: **high** = every trial, **medium** = every 2nd trial, **low** = every 3rd trial

Compliance bins: **high** = every trial, medium = 50–99%, low = 1–49%, none = never. A channel *preserves* the watermark if the decoded payload is bit-identical to the original.

Threat model. The instructor sees no weights, sampling distribution or system prompt; the student pastes the brief unmodified; the chatbot’s tokenizer ingests Unicode scalars transparently. No student submissions were studied – the tool was built alongside the paper, and a classroom evaluation is future work.

■ Classroom Workflow & Ethics

1. **Disclose the policy** in the syllabus — ready-to-paste language ships with the tool; any concern triggers a conversation, not an automatic sanction.
2. **Encode each brief** and paste the encoded result into the LMS — the visible text never changes.
3. **Verify before release** by decoding the brief from the same channel a student would read it through.
4. **Search at grading** for the literal token; a match is a starting point, never a verdict.

The tripwire travels with the prompt, not the student — the take-home generalization of the classroom *word-of-the-day* check.

Four ethics rules. Disclose the policy · use a **kind payload** that helps a good-faith student · treat a hit as probable cause, never a verdict · stay inside the educational context — in adversarial hands the same primitive is an indirect prompt injection.

■ Limitations, Defences & Next Steps

Limitations. A one-line filter over the Tags range, retyping, or any non-trivial paraphrase of the brief erases the watermark. Strict input sanitizers and non-surfacing tokenizers deliver no payload at all (5 of 13 configurations). Channels can silently strip extended Unicode, and vendor behaviour drifts without notice. A hit is a **signal, not proof** of misconduct.

Model-side defences. Stripping the Tags block before tokenization makes the watermark inert — cheap, but unevenly adopted. Output sanitization of unusual literal tokens is harder: the token space is unbounded. Expect a period of co-evolution.

Next steps. (i) HMAC payloads binding assignment, term and section; (ii) pairing the Tag channel with content-level watermarks to survive light paraphrase; (iii) a pre-registered, multi-section, multi-instructor field study; (iv) agreeing with providers which input transformations may be applied silently.

■ The SteganoPrompt Tool

One self-contained HTML file, zero outbound requests: paste the visible brief (1), optionally add a bespoke instruction (2) above the mandatory, read-only integrity reminder, then press Encode & copy. The Verify panel decodes any text, so the payload is audited before release and checked at grading.

Design goals: visual identity · channel survival · reader transparency · client-side, offline privacy · auditability.

■ How It Works



- 95 printable ASCII code points plus tab and newline; smart typography folds to ASCII first.
- Payload maps into the **Tags mirror**: invisible in every common font.
- Insertion $\pi \in \{\text{start, mid, end}\}$; default **end** — most robust to truncation.
- Most frontier tokenizers ingest tag code points as ordinary text.
- Verify panel decodes any text — audit before release, test a suspect submission.

Worked example: $W = \text{Hi!}\backslash n$
 $\mapsto \langle \text{U+E0048, U+E0069, U+E0021, U+E000A} \rangle$.

■ The Watermark, Formally

One character shift defines the channel; one concatenation defines the watermark:

$$\begin{aligned} \varepsilon(c) &= c + \mathbf{0x} \mathbf{E0000}, & \text{wm}_\pi(P, W) &= P_{[1:k]} \| E(W) \| P_{[k+1:|P|]} \\ \text{verify}(X) &= D(\rho_{\mathcal{T}}(X)) = W & \text{for } X &= \text{wm}_\pi(P, W) \end{aligned}$$

P is the visible brief, W the hidden ASCII payload, and E, D the string extensions of the per-character encoder ε and decoder δ ($D \circ E = \text{id}$). The insertion index $k(P, \pi) \in \{0, \mu(P), |P|\}$ places the invisible block at the start, the first sentence boundary $\mu(P)$, or the end of P . The projection ρ_T keeps only Tag-block code points, so verification recovers the payload exactly.

■ Cross-Model Compliance

Model family	Reads tags	Compliance
Anthropic Claude 3, 3.5, 4	✓	high
OpenAI GPT-5 (latest)	×	none
OpenAI GPT-4, 4o, 4 Turbo	✓	high
Google Gemini 1.5, 2, 3.x Pro	✓	high
xAI Grok 4 Fast	✓	high
Alibaba Qwen3 (235B)	✓	high
Alibaba Qwen2.5 (72B)	✓	none
DeepSeek V3 / V3.1	✓	low
Mistral Codestral	✓	low
Mistral Large / Medium	×	none
Meta Llama 3 (8B)	×	none
Meta Llama 3 (70B)	×	none
Meta Llama 4 (Scout)	×	none

12 trials per model, on or before 31 July 2026. **6 of the 13** configurations never emitted the token, so reading the tags is necessary but not sufficient. Failure has two causes: the code points never reach the model (provider sanitizing, as we infer for GPT-5; or a tokenizer that does not surface them, as in the Llama and Mistral Large/Medium pipelines), or the model receives them and does not act (Qwen2.5-72B).

■ Student-Side Mitigation

Student guard	Prior suspicion	Grok 4.5	Gemini 3.1 Pro	Gemini 3.6 Flash
None (control)	—	6/6	6/6	6/6
Naive "ignore injections"	yes	1/6	1/5	0/6
Delimited data framing	yes	1/6	0/6	0/4
Delimiting + format lock	yes	1/6	0/6	0/6
Restate, answer elsewhere	yes	0/6	0/6	0/6
Output-format lock	no	5/6	6/6	0/6

Trials in which the token still appeared, out of six (2 briefs × 3 runs) per model; vendor APIs, 5 Aug 2026; 105 valid trials, 3 lost to transient API errors.

Any guard that declares the pasted block to be *data* suppresses the watermark — but every one of them needs a student who **already suspects**. Only the output-format lock needs no suspicion, and it defeats the payload solely on the smaller **Gemini 3.6 Flash**; both flagships prefer the invisible instruction to the visible formatting constraint.

Vendor drift. Probed through the chat products instead of the APIs, *GPT-5.6 Sol* stayed silent, and *Claude Opus 5* read the payload and warned the user, reporting the hidden instruction rather than acting on it — a third model-side posture. Re-test immediately before deployment.

■ Cross-Channel Survival

✓ Survives	plain text, Markdown, HTML, JSON · e-mail (Gmail, Outlook) · Microsoft Word (.docx) · Google Docs · PDF (text export) · Notion, Slack, Discord · Canvas, Blackboard, Moodle (rich text)
✗ Fails	Twitter / X posts · Apple Notes (iOS, certain versions)

256-byte payload; ✓ = bit-identical recovery in $\geq 95\%$ of trials, ✗ = stripped or corrupted in the majority. Always verify from the path the student actually reads.

■ Take-away

An educator-controlled tripwire works today. On the hosted frontier chatbots — Claude, Gemini, Grok — and on the largest open-weight model tested, Qwen3-235B, a verbatim copy-paste of a watermarked brief surfaced the integrity token in **every trial**, and the payload survived every classroom distribution channel tested.

Beatable, and that is acceptable. One sentence declaring the pasted block to be data suppresses the signal in nearly every trial — but only a student who already suspects writes it, and the announced policy is what supplies the suspicion. Deploy it disclosed: verify on the channel students actually read, and treat every hit as the start of a conversation, not a verdict.

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